FPV DRONE RACING

PROJECT NAME:	FPV Drone Racing – 3D Printed Propeller Guard
PROJECT GROUP:	CLASS OF 2024

WHY THIS PROJECT ?

- FPV racing drones operate at high speed, increasing crash risk.
- Propellers are the most damage-prone and dangerous components.
- Crashes can cause injury, propeller breakage, and motor damage.
- Commercial propeller guards are costly and not customizable.
- 3D printing allows lightweight, low-cost, and replaceable safety guards.

INTRODUCTION

- FPV drone racing requires high agility and rapid maneuvers, often leading to collisions.
- The 3D Printed Propeller Guard is designed to protect propellers, motors, and surroundings without affecting flight performance significantly.
- The guard is custom-designed and fabricated using 3D printing, allowing easy modification.
- It improves safety during training, beginner practice, and close-proximity racing.

KEY FEATURES

- Lightweight 3D printed design
- Protects propellers from direct impact
- Reduces injury risk to pilots and spectators
- Custom-fit for FPV racing frames
- Easy installation and replacement
- Improves durability of racing drone

ARCHITECTURE

- The flight controller acts as the main control unit of the drone.
- The power system supplies energy to all electronic components.
- Motors and propellers generate lift and movement.
- The radio system allows the pilot to control the drone wirelessly.
- The FPV system provides live video transmission to the pilot.

RESULTS

- The FPV drone successfully flew with the 3D printed propeller guard installed.
- The guard provided effective protection during minor crashes.
- Minimal impact on flight stability and maneuverability.
- Improved confidence during training sessions.

FUTURE SCOPE

1. **Use of flexible TPU filament for better shock absorption**
2. **Aerodynamic optimization to reduce drag**
3. **Integration with racing frames**